

$$\text{Si } f(x) = \ln u$$

ENTONCES:

$$f'(x) = \frac{u'}{u}$$

$$y = \ln(\underline{3x}) \text{ Si } f(x) = \ln u$$

$$y' = \frac{\cancel{3}}{\cancel{3}x}$$

$$y' = \frac{1}{x}$$

ENTONCES:

$$f'(x) = \frac{u'}{u}$$

$$f(x) = \ln(\underline{5x^2})$$

$$f'(x) = \frac{\cancel{10}x^2}{\cancel{5}x^2}$$

$$f'(x) = \frac{2}{x}$$

$$\text{Si } f(x) = \ln u$$

ENTONCES:

$$f'(x) = \frac{u'}{u}$$

$$y = 3 \cdot \ln(\underline{2x^3})$$

$$y' = \frac{3}{1} \cdot \frac{\cancel{6}x^3}{\cancel{2}x^3}$$

$$y' = \frac{9}{x}$$

$$f(x) = 2 \ln x^3$$

$$y = \ln(3x^2 - 2x)$$

51 $f(x) = e^x$

$$f'(x) = e^x$$

$$f(x) = e^{3x}$$

$$f'(x) = e^{3x} \cdot 3$$

$$f'(x) = 3e^{3x}$$

51 $f(x) = e^x \in \mathbb{R}$

$$f'(x) = e^x$$

$$f(x) = e^{x^3}$$

$$f'(x) = e^{x^3} \cdot 3x^2$$

$$f'(x) = 3x^2 \cdot e^{x^3}$$

$$f(x) = e^{2x^2}$$

$$f(x) = e^{x^2+2x}$$

$$\text{Si } f(x) = \text{Sen } x$$

$$f'(x) = \text{Cos } x$$

$$f(x) = \text{Sen } 3x$$

$$f'(x) = \text{Cos } 3x \cdot 3$$

$$f'(x) = 3 \text{Cos } 3x$$

$$\text{Si } f(x) = \text{Sen } x \text{ ENTON}$$

$$f'(x) = \text{Cos } x$$

$$f(x) = \text{Sen}^2 x$$

$$f(x) = (\text{Sen } x)^2$$

$$f'(x) = 2(\text{Sen } x)' \cdot \text{Cos } x$$

$$f'(x) = 2 \cdot \text{Sen } x \cdot \text{Cos } x$$

$$f(x) = \text{Sen } x^3$$

$$f(x) = \text{Sen}^3 x$$

$$f(t) = \text{Sen}(4t^2 + t)$$

$$y = \text{Sen}^3(2x)$$

$$f(x) = \underbrace{\text{Sen}}_{(1)}(\underbrace{3x^2 - 5}_{(2)})$$

$$f'(x) = \underbrace{\text{Cos}(3x^2 - 5)}_{(1)} \cdot \underbrace{6x}_{(2)}$$

$$f'(x) = 6x \cdot \text{Cos}(3x^2 - 5)$$

$$y = \text{Sen}^5(3x^3 + 2x)$$

$$y = x$$

$$y' = 5x^4$$

$$y = \left[\underbrace{\text{Sen}}_{(2)}(\underbrace{3x^3 + 2x}_{(3)}) \right]^{(1)}$$

$$y' = \underbrace{5}_{(1)} \left[\underbrace{\text{Sen}(3x^3 + 2x)}_{(1)} \right]^4 \cdot \underbrace{\text{Cos}(3x^3 + 2x)}_{(2)} \underbrace{(9x^2 + 2)}_{(3)}$$

$$y' = (45x^2 + 10) \text{Sen}^4(3x^3 + 2x) \cdot \text{Cos}(3x^3 + 2x)$$

MA

$$\text{Si } f(x) = \cos x \text{ EN}$$

$$f'(x) = -\text{Sen } x$$

$$f(x) = \cos 2x$$

$$f'(x) = -\text{Sen } 2x \cdot 2$$

$$f'(x) = -2 \text{ Sen } 2x$$

$$f(x) = \cos^2 x$$

$$f(x) = (\cos x)^2$$

$$f'(x) = 2(\cos x)' \cdot (-\text{Sen } x)$$

$$f'(x) = -2 \text{ Sen } x \cdot \cos x$$

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$$f(x) = \cos 3x^2$$

$$f(x) = \cos^3 x$$

$$f(u) = \cos(u^5 - 3)$$

$$y = \cos^5(x^2 + x)$$

$$y = \underbrace{\cos}_{(1)}(\underbrace{5x^3 - 3x^2}_{(2)})$$

$$y' = \underbrace{-\text{Sen}(5x^3 - 3x^2)}_{(1)} \cdot \underbrace{(15x^2 - 6x)}_{(2)}$$

$$y' = -(15x^2 - 6x) \text{ Sen}(5x^3 - 3x^2)$$

$$f(x) = \cos^3(2x^2 + 3x)$$

$$f(x) = [\cos(2x^2 + 3x)]^3$$

$$f'(x) = \underbrace{3[\cos(2x^2 + 3x)]^2}_{(1)} \underbrace{[-\text{Sen}(2x^2 + 3x)]}_{(2)} \underbrace{(4x + 3)}_{(3)}$$

$$f'(x) = -(12x + 9) \cdot \text{Sen}(2x^2 + 3x) \cdot \cos^2(2x^2 + 3x)$$

$$d) f(x) = \tan x$$

$$f'(x) = \sec^2 x$$

$$f(x) = \tan 3x$$

$$f'(x) = \sec^2 3x \cdot 3$$

$$f'(x) = 3 \sec^2 3x$$

$$f(x) = \tan^2 x$$

$$f(x) = (\tan x)^2$$

$$f'(x) = 2(\tan x)' \cdot \sec^2 x$$

~~$$f'(x) = 2 \tan x \sec^2 x$$~~

$$f(x) = \tan x^2$$

$$f(x) = \tan^3 x$$

$$f(x) = \tan^5(x^2 + 2x)$$

$$y = \tan(7x^6 - 5x^3 + 8)$$

$$y = \tan(\underbrace{5x^2 + x}_{(2)})$$

$$y' = \underbrace{\sec^2(5x^2 + x)}_{(1)} \cdot \underbrace{(10x + 1)}_{(2)}$$

$$y' = (10x + 1) \sec^2(5x^2 + x)$$

$$f(x) = \left[\underbrace{\tan}_{(2)}(\underbrace{2x^3 + 5}_{(3)}) \right]^{\overset{(1)}{3}}$$

$$f'(x) = \underbrace{3 \left[\tan(2x^3 + 5) \right]^2}_{(1)} \cdot \underbrace{\sec^2(2x^3 + 5)}_{(2)} \cdot 6x^2$$

$$f'(x) = 18x^2 \cdot \tan^2(2x^3 + 5) \cdot \sec^2(2x^3 + 5)$$

$$f(x) = \cot x \quad \text{ENT}$$

$$f'(x) = -\csc^2 x$$

$$f(x) = \cot 2x$$

$$f'(x) = -\csc^2 2x \cdot 2$$

$$f'(x) = -2 \csc^2 2x$$

$$f'(x) = -\csc^2 x$$

$$f(x) = \cot^2 x$$

$$f(x) = (\cot x)^2$$

$$f'(x) = 2(\cot x)' \cdot (-\csc^2 x)$$

$$f'(x) = -2 \cot x \cdot \csc^2 x$$

$$f(x) = \cot x^2$$

$$f(x) = \cot^3 x$$

$$f(x) = \cot^6(x^3 - x^2 + 5)$$

$$y = \cot(5x^2 + 6x - 3)$$

$$y = \cot(3x^4 - 2x^3)$$

$$y' = -\csc^2(3x^4 - 2x^3) \cdot (12x^3 - 6x^2)$$

$$y' = -(12x^3 - 6x^2) \cdot \csc^2(3x^4 - 2x^3)$$

$$f(x) = \cot^4(2x^2 - 5)$$

$$f(x) = [\cot(2x^2 - 5)]^4$$

$$f'(x) = 4[\cot(2x^2 - 5)]^3 \cdot [-\csc^2(2x^2 - 5)] \cdot 4x$$

$$f'(x) = -16x \cdot \cot^3(2x^2 - 5) \cdot \csc^2(2x^2 - 5)$$

$$f(x) = \sec x \text{ ENTONCES}$$

$$f'(x) = \sec x \cdot \tan x$$

$$f(x) = \sec 3x$$

$$f'(x) = \sec 3x \cdot \tan 3x \cdot 3$$

$$f'(x) = 3 \sec 3x \cdot \tan 3x$$

$$f(x) = \sec^2 x$$

$$f(x) = (\sec x)^2$$

$$f'(x) = 2(\sec x)' \cdot \sec x \tan x$$

$$\cancel{f'(x) = 2 \sec^2 x \cdot \tan x}$$

$$f(x) = \sec x^3$$

$$f(x) = \sec^3 x$$

$$y = \sec^2(-3x^2 + x + 1)$$

$$f(x) = \sec(2x - 4)$$

$$y = \underbrace{\sec}_{(1)}(\underbrace{3x^2 + 2x}_{(2)})$$

$$y' = \underbrace{\sec(3x^2 + 2x) \cdot \tan(3x^2 + 2x)}_{(1)} \cdot \underbrace{(6x + 2)}_{(2)}$$

$$y' = (6x + 2) \sec(3x^2 + 2x) \cdot \tan(3x^2 + 2x)$$

$$f(x) = \sec^4(x^5)$$

$$f(x) = \left[\underbrace{\sec}_{(2)} \underbrace{x^5}_{(3)} \right]^{(4)}$$

$$f'(x) = 4 \left[\sec x^5 \right]^3 \cdot \left[\sec x^5 \right]' \tan x^5 \cdot 5x^4$$

$$f'(x) = 20x^4 \cdot \sec^3 x^5 \cdot \sec' x^5 \cdot \tan x^5$$

$$f'(x) = 20x^4 \cdot \tan x^5 \cdot \sec^4 x^5$$

$$f(x) = \csc x \quad \text{ENTON}$$

$$f'(x) = -\csc x \cdot \cot x$$

$$f'(x) = \csc 5x$$

$$f'(x) = \csc 5x$$

$$f'(x) = -\csc 5x \cdot \cot 5x \cdot 5$$

$$f'(x) = -5 \csc 5x \cdot \cot 5x$$

$$f(x) = \csc^2 x$$

$$f(x) = (\csc x)^2$$

$$f'(x) = 2(\csc x)' \cdot (-\csc x \cdot \cot x)$$

$$f'(x) = -2 \csc^2 x \cdot \cot x$$

$$f(x) = \csc x^3$$

$$f(x) = \csc^3 x$$

$$y = \csc (x^3 + x^2 - x)$$

$$f'(x) = \csc^3 (3x^2 + 2)$$

$$y = \csc (4x^6 - 3x^5)$$

$$y' = -\csc (4x^6 - 3x^5) \cdot \cot (4x^6 - 3x^5) (24x^5 - 15x^4)$$

$$y' = -(24x^5 - 15x^4) \cdot \cot (4x^6 - 3x^5) \cdot \csc (4x^6 - 3x^5)$$

$$f(x) = \csc^4 (x^2 - 2)$$

$$f(x) = [\csc (x^2 - 2)]^4$$

$$f'(x) = 4 [\csc (x^2 - 2)]^3 \cdot [-\csc (x^2 - 2) \cdot \cot (x^2 - 2)] \cdot 2x$$

$$f'(x) = -8x \cdot \csc^3 (x^2 - 2) \cdot \csc (x^2 - 2) \cdot \cot (x^2 - 2)$$

$$f'(x) = -8x \cdot \csc^4 (x^2 - 2) \cdot \cot (x^2 - 2)$$